

Coding & Solution

Date	02 October 2026
Team ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python (backend), HTML, CSS and JavaScript (frontend)
Framework(s) Used	FastAPI with Uvicorn (ASGI server), Jinja2 templates, Pydantic; Google Gen AI SDK (google-genai) for Gemini; optional Transformers + PyTorch for the local LaMini-Flan-T5-783M model
Key Features Implemented	• Question answering (/qa) with simple, careful explanations • Concept explanation (/explain) for beginner, intermediate or advanced level; optional local LaMini-Flan-T5-783M with automatic Gemini fallback • Interactive 3-question multiple-choice quiz (/quiz) with instant correct / incorrect feedback • Summarization of study notes (/summarize) • Personalized learning path with ordered topics, time estimates and prerequisites (/learn/recommendations) • Web interface (HTML + CSS + Jinja2) with task and level selection, copy-result button and a /health endpoint • Input validation and clear error messages
Pending / Incomplete Features	• Diagnostic readiness check and learning-gap detection • Study plan built from deadlines and available time • Progress tracking, goals and dashboard (no database yet) • User authentication, rate limiting and cloud deployment
Setup / Run Instructions	1. Install Python 3.10 or newer and open the EduGenie folder in VS Code. 2. Create and activate a virtual environment (python -m venv .venv). 3. Run: pip install -r requirements.txt 4. Copy .env.example to .env and set GEMINI_API_KEY to your Gemini API key (do not share or commit .env). 5. Start the app: uvicorn main:app --reload 6. Open http://127.0.0.1:8000 (API docs at /docs). Optional: pip install torch transformers and set USE_LOCAL_EXPLAINER=true to use the local explanation model.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The Gemini model is configurable through GEMINI_MODEL in .env (the code defaults to gemini-2.5-flash).
- The local LaMini-Flan-T5-783M explainer is optional and falls back to Gemini if it fails or is disabled.
- AI responses can be inaccurate; students should verify important facts.
- Unit tests for quiz JSON clean-up are included in tests/ (python -m unittest discover -s tests -v).